

TIMOTHY JAMES R. BRANNON

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EDUCATION

Clemson University, Clemson, SC

B.S. Computer Engineering | GPA: 3.26/4.00 | Expected December 2026

Greenville Technical College, Greenville, SC

A.S. | GPA: 3.66/4.00 | December 2023

TECHNICAL SKILLS

Programming: C, C++, MATLAB, Python, SQL, Java

Embedded: ESP32, Arduino, Embedded C/C++, ESP-NOW, I2S, Sensors, Interrupts

Engineering: DC Motor Control, Feedback Control, Computer Vision, Signal Processing

Software: React Native, Expo, PostgreSQL, Supabase, Git, GitHub

Tools: MATLAB App Designer, SolidWorks, Arduino IDE

ENGINEERING PROJECTS

Automated Gameboard — Senior Design I | MATLAB, Motor Control, Computer Vision

- Developed software integrating DC motor control, feedback control, computer vision, game logic, and MATLAB GUI.
- Implemented closed-loop motor control and computer vision to identify dice outcomes and automate game decisions.

IoT Smart Light Detection and Audio Alert System | ESP32, C/C++, ESP-NOW, I2S

- Developed a two-node ESP32 system using light and door-state sensors to detect when a light may have been left on.
- Implemented ESP-NOW communication, embedded decision logic, and audio alerts using I2S, MAX98357A, and flash-based audio storage.

Throne Finder — Mobile Application | React Native, Expo, Supabase, PostgreSQL

- Developing a cross-platform application for locating and rating public restrooms through an interactive map.
- Designing PostgreSQL/Supabase backend with authentication, user data, and row-level security.
- Managing development and version control using Git and GitHub.

EXPERIENCE

Fairway Ford, Greenville, SC

Car Porter | May 2025 – Present

- Coordinate vehicle movement, customer pickup/delivery, and communication between customers and service advisors.
- Manage vehicle transfers while maintaining safety and accuracy.

Chick-fil-A, Taylors, SC

Team Member | August 2023 – April 2024

- Worked in a high-volume environment requiring teamwork, communication, accuracy, and rapid decision-making.

Mast General Store, Greenville, SC

Sales Associate | August 2019 – May 2023

- Trained new employees and developed organizational systems to improve efficiency and reduce errors.

HONORS AND ACTIVITIES

President's List — Greenville Technical College

Dean's List — Multiple Semesters

Volunteer — Samaritan's Purse Tornado Cleanup

Youth Ministry Intern — Rock Springs Baptist Church